BIOS 794: Generalized Additive Models

Alexander McLain

Introduction

- So far, the regression methods that we've used have assumed some linear function $\boldsymbol{X}'\boldsymbol{\beta}$.
- In this section, we explore non-linear functions $h_m(\boldsymbol{X}) : \mathbb{R}^p \to \mathbb{R}$ for $m = 1, \ldots, M$ and use the model
$$f(\boldsymbol{X}) = \sum_{m=1}^{M} \beta_m h_m(\boldsymbol{X})$$
- The $h_m(\boldsymbol{X})$ we'll consider are *basis expansions*.
- There are many types of **basis expansions**: *polynomials, piecewise-polynomials, splines,* and *wavelets* are a few.

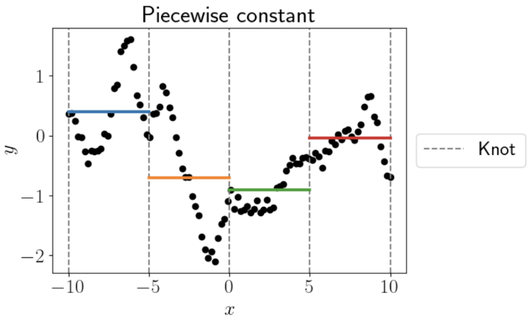
Piecewise-polynomials and splines

- For the moment, we'll consider one-dimensional $\boldsymbol{X}$, e.g., we don't know the impact of BMI and 6-month change in HbA1C.
- Piecwise constant with 2 knots

$$h_1(X) = I(X < \xi_1)$$

$$h_2(X) = I(\xi_1 \leq X < \xi_2)$$

$$h_3(X) = I(X > \xi_2)$$



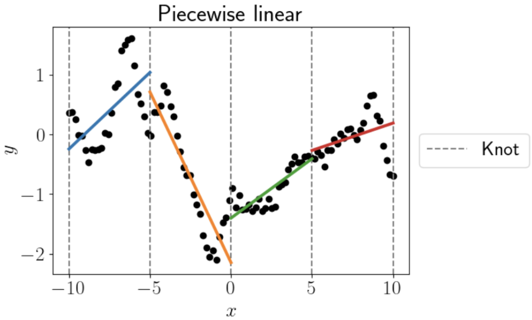
Piecewise-polynomials and splines

- Piecwise linear with 2 knots would have everything from the piecewise constant plus:

$$h_4(X) = X \times I(X < \xi_1)$$

$$h_5(X) = X \times I(\xi_1 \leq X < \xi_2)$$

$$h_6(X) = X \times I(X > \xi_2)$$



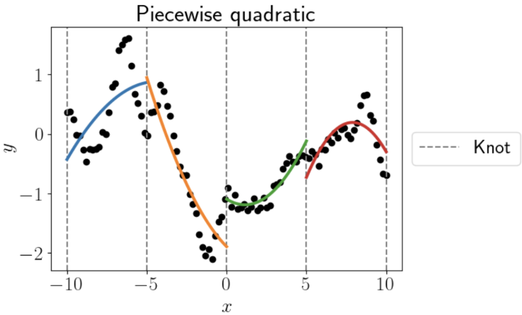
Piecewise-polynomials and splines

- Piecwise quadratic with 2 knots would have everything from the piecewise linear plus:

$$h_7(X) = X^2 I(X < \xi_1)$$

$$h_8(X) = X^2 I(\xi_1 \leq X < \xi_2)$$

$$h_9(X) = X^2 I(X > \xi_2)$$

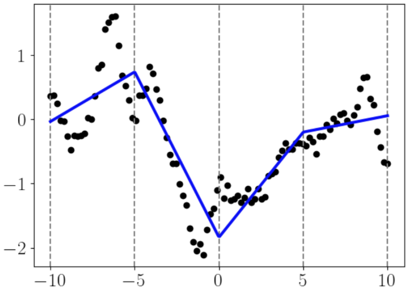


Piecewise-polynomials and splines

- The previous versions have discontinuities.
- Continuous piecewise linear version

$$h_1(X) = 1 \quad h_2(X) = X \quad h_3(X) = (X - \xi_1)_+$$

- Note that the first derivative (i.e., the slope) is not continuous.



Piecewise Cubic Polynomials

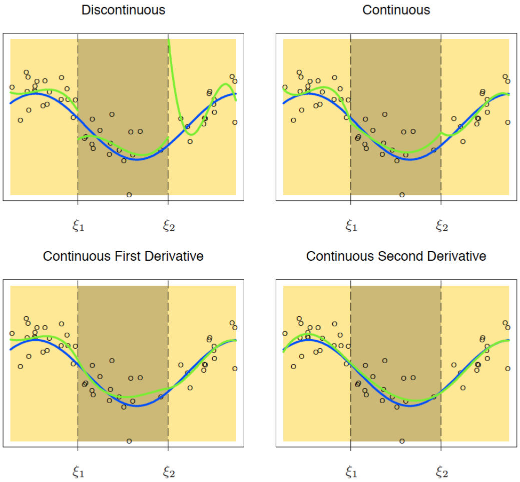


Figure: From ESL (online version).

Cubic splines

- Having a continuous second derivative leads to *cubic spline*
  - (note: continuous second derivative functions are smooth to the naked eye, going beyond second doesn't make them look more smooth)
- For our example we have

$$h_1(X) = 1 \quad h_3(X) = X^2 \quad h_5(X) = (X - \xi_1)_+^3$$

$$h_2(X) = X \quad h_4(X) = X^3 \quad h_6(X) = (X - \xi_2)_+^3$$

- The number of parameters is: (3 regions)$\times$(4 parameters per region) - (2 knots)$\times$(3 constraints per knot) $= 6$.

order-M splines

- An order-M spline (or a *piecewise-polynomial of order M*) has continuous derivatives to up to order $M-2$ (for cubic spline $M=4$).
- With knots at ξ_j for $j=1,\dots,K$ we have

$$\begin{aligned} h_j(X) &= X^{j-1} \quad \text{for } j=1,\dots,M \\ h_{M+\ell}(X) &= (X-\xi_\ell)_+^{M-1}, \quad \text{for } \ell=1,\dots,K \end{aligned}$$

- These **fixed knot splines** are known as *regression splines*

Natural Cubic Splines

- Polynomials are erratic towards the boundary of the data.
- This can be exacerbated with piecewise-polynomial splines.
- *Natural cubic splines* add the constraint that the function is linear beyond the boundary of the data.
- This constraint decreases the *degrees of freedom* of the model (i.e., we could add more interior knots) and adds stability to the model at and beyond the boundary.
    - it also may add bias, but some bias would be preferred since we don't have much data in this region.

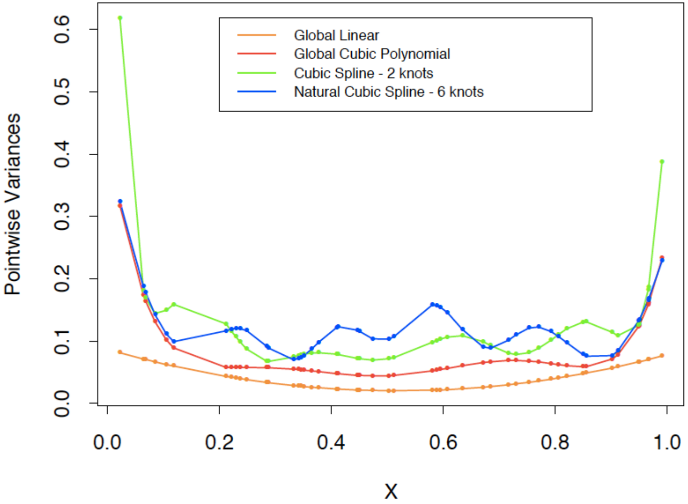


Figure: Pointwise variance of different spline methods. From ESL (online version) page 145.

Natural Cubic Splines

- Natural Cubic Splines with K knots are represented by K basis functions.
- They can be derived by reducing the basis of cubic spline by imposing the boundary constraints.
- In this case they take the form

$$N_1(X) = 1, \quad N_2(X) = X, \quad N_{k+2}(X) = d_k(X) - d_{K-1}(X),$$

where

$$d_k(X) = \frac{(X - \xi_k)^3_+ - (X - \xi_K)^3_+}{\xi_K - \xi_k}.$$

- Note that each of these basis functions has zero 2nd and 3rd derivative outside the boundary knots

Cubic B-Splines

- The previous splines we've discussed are good to learn from since they have relatively simple forms.
- In practice, (by far) the most used spline is the B-spline.
- Let $B_{i,m}(x)$ be the ith B-spline basis function of order m with knot sequence τ.
- The B-splines are defined recursively as $B_{i,1} = I(\tau_i \leq x < \tau_{i+1})$ for $i = 1, \ldots, K + 2M - 1$ and

$$B_{i,m}(x) = \frac{x - \tau_i}{\tau_{i+m-1} - \tau_i} B_{i,m-1}(x) + \frac{\tau_{i+m} - x}{\tau_{i+m} - \tau_{i+1}} B_{i+1,m-1}(x)$$

- The most common is the $m = 4$ cubic B-splines. There are also natural cubic B-splines.

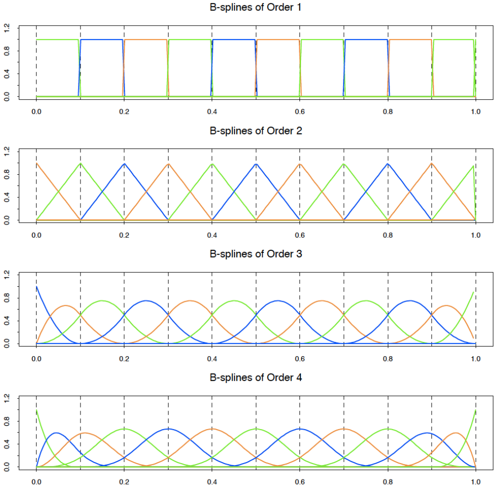


Figure: B-splines or varying order. From ESL (online version) page 188.

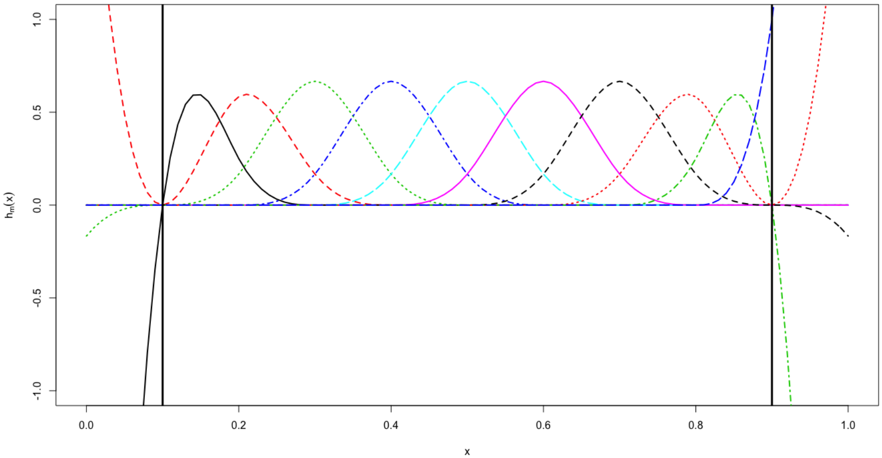


Figure: Cubic B-splines with $df = 10$ and boundary knots at 0.1 and 0.9.

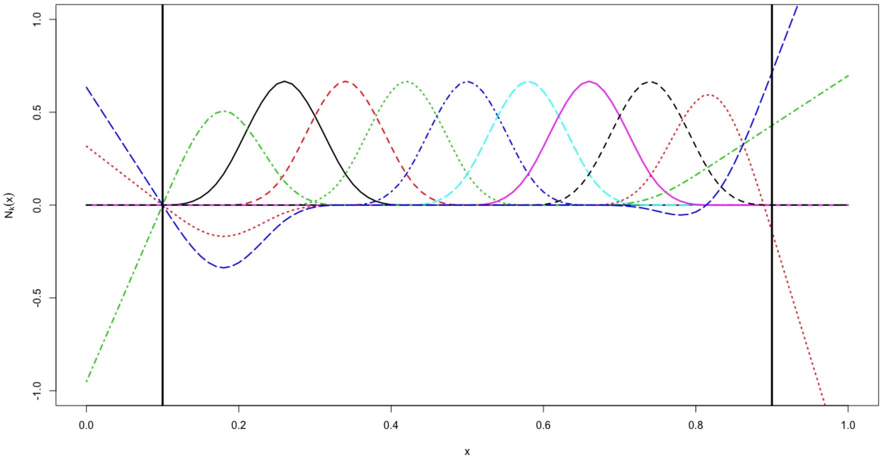


Figure: Natural cubic B-splines with $df = 10$ and boundary knots at 0.1 and 0.9.

Introduction

- The previous slides we had to choose the number and location of the interior knots of the spline function.
- To avoid the knot selection issue we can use the penalized RSS

$$RSS(f,\lambda) = \sum_{i=1}^{N} \{y_i - f(x_i)\}^2 + \lambda \int \{f''(t)\}^2 dt$$

  where λ is the *smothing parameter*
- Some special cases:
  - $\lambda = 0$: f interpolates the data
  - $\lambda = \infty$: reverts to the LS fit wrt x with no second derivative

Smoothing Spline Solution

- Interestingly, it can be shown that $RSS(f, \lambda)$ has an explicit, finite-dimensional, unique minimizer which is a *natural cubic spline* with knots at **the unique values of the x_i for $i = 1, \ldots, N$.**
- While having (potentially) N knots might seem crazy the penalty term will shrink the spline coefficients to zero.
- We write the solution to this model as

$$f(x) = \sum_{i=1}^{N} N_j(x)\theta_j$$

where $N_j(x)$ represent the basis functions of this family of natural splines.

Smoothing Spline Solution

- We can re-write the penalized RSS as

$$RSS(f,\lambda) = (\boldsymbol{y} - \boldsymbol{N}\theta)'(\boldsymbol{y} - \boldsymbol{N}\theta) + \lambda\theta'\Omega_N\theta$$

  where $\boldsymbol{N}_{ij} = N_j(x_i)$ and $\{\Omega_N\}_{jk} = \int N_j''(t)N_k''(t)dt$.
- Note the similarities of this RSS and ridge regression. If fact, this is known as a *generalized ridge regression*
- The solution is given by

$$\hat{\theta} = (\boldsymbol{N}'\boldsymbol{N} + \lambda\Omega_N)^{-1}\boldsymbol{N}'\boldsymbol{y}$$

Selecting λ via degrees of freedom

- We can estimate the *effective degrees of freedom* via

$$df_\lambda = \mathrm{trace}(\boldsymbol{S}_\lambda)$$

  where $\boldsymbol{S}_\lambda$ is known as the smoother matrix.
- This gives us a way to specify the amount of smoothing in terms of the degrees of freedom (i.e., number of parameters).
- For example, we can set λ by solving $df_\lambda = 12$ for λ.
- The package that we'll use selects λ using REML.

Introduction

- So far we have focused on one-dimensional spline models. Each of the approaches have multidimensional analogs.
- Suppose $X \in \mathbb{R}^2$ and we have univariate function $h_{1j}(X_1)$ for $j = 1, \ldots, M_1$ and $h_{2j}(X_2)$ for $j = 1, \ldots, M_2$.
- Then the $M_1 \times M_2$ dimensional tensor product basis defined by

$$g_{jk}(X) = h_{1j}(X_1)h_{2k}(X_2) \; j = 1, \ldots, M_1 \;\; k = 1, \ldots, M_2$$

gives the two-dimensional function

$$g(X) = \sum_{j=1}^{M_1} \sum_{k=1}^{M_2} \theta_{jk} g_{jk}(X).$$

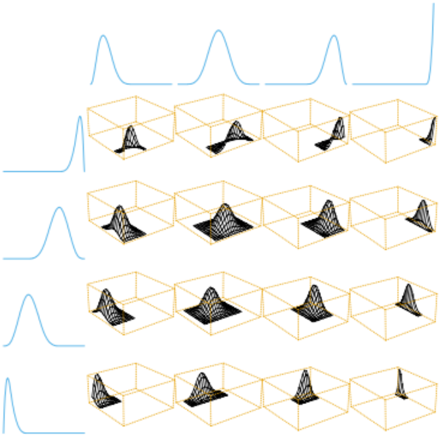


FIGURE 5.10. *A tensor product basis of B-splines, showing some selected pairs. Each two-dimensional function is the tensor product of the corresponding one dimensional marginals.*

Figure: A tensor product of basis of B-splines. From ESL (online version) page 163.

Penalization

- The above can be generalized to d dimensions, but the number of parameters grows exponentially (curse of dimensionality).
- Tensor basis functions can again be fitted with a penalized RSS

$$\sum_{i=1}^{N}\{y_i - f(x_i)\}^2 + \lambda J[f]$$

where J is a penalty functional for the f, which is no longer univariate.

Introduction and motivation

- The previous used a few spline bases function to represent smooth functions.
- Wavelets can be used to represent a mostly flat function with a few isolated bumps.
- Wavelet basis can be thought of as bumpy basis functions.
- They differ from previous basis in that they have both *time and frequency* localization

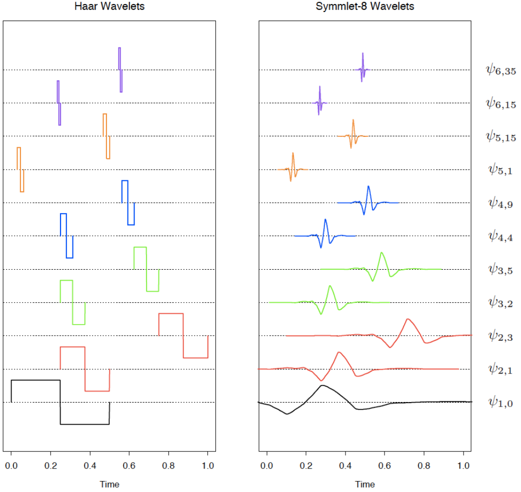


Figure: Harr and Symmlet-8 wavelets. From ESL (online version) page 175.

Adaptive Wavelet Filtering

- A popular method for adaptive wavelet fitting is known as SURE shrinkage (Stein Unbiased Risk Estimation, Donoho and Johnstone (1994)).
- This starts with the lasso criteria

$$\min_{\theta} ||\boldsymbol{y} - \boldsymbol{W}\theta||_2^2 + 2\lambda||\theta||_1,$$

  then since $\boldsymbol{W}$ is orthonormal the solution is

$$\hat{\theta}_j = \text{sign}(y_j^*)(|y_j^*| - \lambda)_+$$

- The interesting thing about wavelets is that coefficients represent characteristics of the signal **localized in time** (the basis functions at each level are translations of each other) and **localized in frequency.**
- Modern image compression is often performed using two-dimensional wavelet representations.

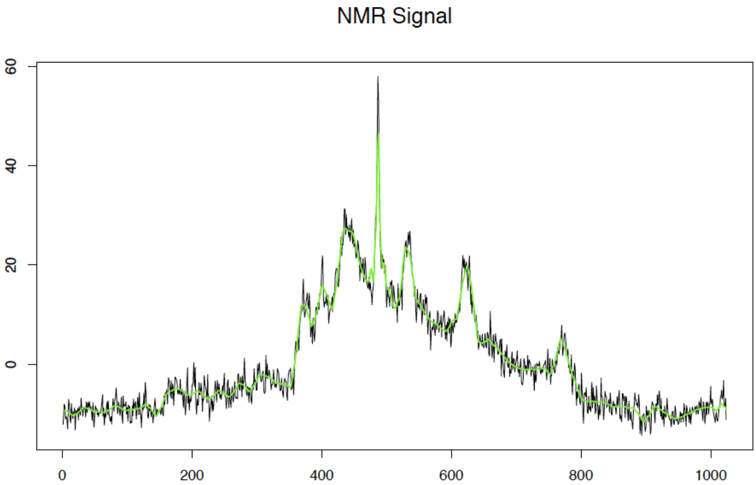


Figure: Truncated and untruncated coefficients of Symmlet-8 wavelets used to de-noise on a nuclear magnetic resonance signal. From ESL (online version) page 175.

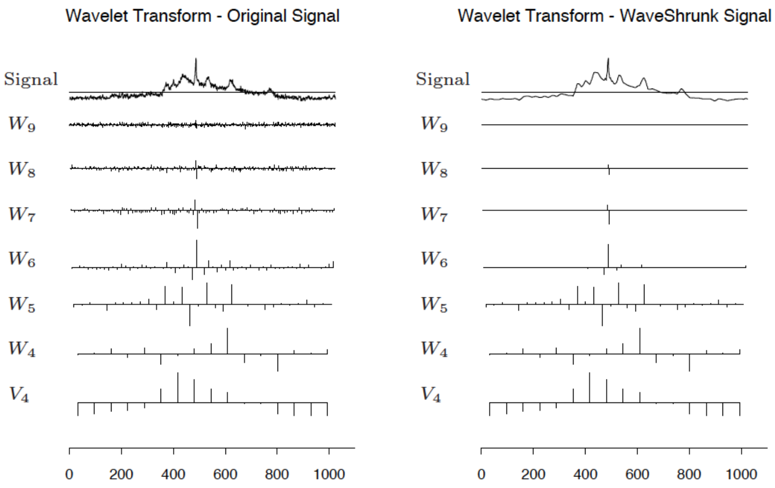


Figure: Example of using Symmlet-8 wavelets to de-noise on a nuclear magnetic resonance signal. From ESL (online version) page 175.

Basis Functions in Practice: Why Use Them?

- **Check Relationships:** I often use basis functions to see whether the relationship between a predictor and outcome is really linear.
- **Capture Curves:** Expansions (like polynomials or splines) let us see curved or more complex patterns that a straight line would miss.
- **Flexible but Simple:** They still use the linear regression framework, so fitting and interpreting them is familiar.
- **Better Predictions:** If a predictor has a non-linear effect, including basis functions can improve model fit and prediction.

Basis Functions in Practice: Things to Watch For

- **Overfitting:** Adding too many terms can chase noise instead of the real signal.
- **Interpretability:** Higher-order terms or spline bases can make interpretation harder.
- **Correlation Among Terms:** Polynomial terms can be highly correlated with each other.
- **Choices Matter:** You must decide how many terms to add (e.g., polynomial degree, number of spline knots).
- **Edge Behavior:** Some basis expansions behave strangely at the extremes of the data.